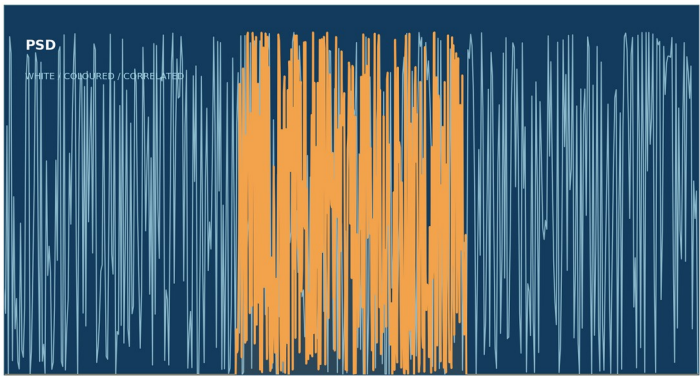


TECHNICAL SCHOOL OF TELECOMMUNICATIONS
ENGINEERING AND SYSTEMS

PRACTICE 05

NOISE
GENERATION

White Gaussian noise, spectral
colouring and statistical analysis



MATLAB LABORATORY REPORT

MATHEMATICAL SIMULATION AND STATISTICAL ANALYSIS OF
THERMAL NOISE IN COMMUNICATION SYSTEMS

STUDENT	INSTITUTION	ACADEMIC YEAR
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Scope of the report: generation of a complex random spectrum, conversion to real Gaussian-like noise, PSD normalization, correlation analysis, FIR low-pass colouring, IIR band-pass colouring, ideal FFT masking and power verification.

1. Introduction and objectives

Thermal noise is one of the fundamental limitations of communication receivers. In an ideal model it is represented as a zero-mean Gaussian random process with a flat power spectral density. Practical systems then shape this process through amplifiers, channels and filters, producing coloured noise whose samples are correlated in time.

This practice implements two complementary MATLAB workflows. The first script synthesizes a white process from a constant-magnitude spectrum with random phase, separates two real realizations and applies a finite-impulse-response low-pass filter. The second script generates band-pass coloured noise with both a fourth-order Butterworth filter and an ideal frequency-domain mask.

1.1 Objectives

- Generate a reproducible white Gaussian-like random sequence from a random-phase spectrum.
- Normalize the FFT so that the one-sided PSD is expressed in W/Hz and integrates to the measured signal power.
- Use auto- and cross-correlation to distinguish white noise from spectrally coloured noise.
- Compare practical FIR/IIR filtering with ideal frequency-domain masking.
- Validate the simulated output power against N_0 multiplied by bandwidth or equivalent noise bandwidth.
- Deliver executable .m scripts and companion .mat files containing the source code as text.

1.2 Software and assumptions

The implementation requires MATLAB and Signal Processing Toolbox. The reference impedance converts voltage squared into watts. Both scripts use fixed random seeds so that figures and printed power estimates can be reproduced. A new seed may be selected to obtain another statistically equivalent realization.

Parameter	Script 1	Script 2
Samples N	1024	2048
Sampling frequency	1024 Hz	512 Hz
Reference impedance	50 ohm	1 ohm
Target one-sided PSD N_0	1e-6 W/Hz	5e-6 W/Hz
Colouring method	30th-order FIR low-pass	4th-order Butterworth and ideal FFT mask

2. Theoretical background

2.1 Noise power and power spectral density

For a one-sided white-noise PSD N_0 extending from 0 Hz to the Nyquist frequency $f_s/2$, the total mean power is

$$P_n = N_0 * (f_s/2)$$

When the samples represent a voltage across an impedance Z , the mean electrical power is estimated as $\text{mean}(x^2)/Z$. The FFT implementation first calculates power per DFT bin and then divides by the bin width $df = f_s/N$. Doubling the positive-frequency bins, except direct current, produces a one-sided PSD.

$$\text{PSD}[k] = 2 * |\text{FFT}\{x\}[k]/N|^2 / (Z * df), \quad 1 \leq k < N/2$$

2.2 Random-phase spectral synthesis

The magnitude of every two-sided spectral sample is fixed to K , while its phase is uniformly distributed between 0 and 2π . The inverse FFT sums many phasors. By the central-limit effect, each time-domain component approaches a Gaussian distribution even though the spectral magnitude is deterministic.

$$S[k] = K \exp(j \phi_k), \quad \phi_k \sim \text{Uniform}(0, 2\pi)$$

$$K = \sqrt{P_n * Z * N}$$

2.3 Correlation and colour

Ideal white noise has an impulse-like autocorrelation: the zero-lag value contains the variance and nonzero lags tend toward zero. Filtering changes this property. A low-pass or band-pass output has a wider autocorrelation function whose oscillation and decay are related to the selected passband. The real and imaginary realizations remain approximately uncorrelated because their cross-correlation fluctuates around zero.

A finite record never produces a perfectly flat PSD or zero correlation at every nonzero lag. The expected behavior is assessed from the overall trend and from integrated power, not from any single spectral sample.

3. White Gaussian noise generation

3.1 Parameter definition and scaling

In `RuidoBlancoGaussiano.m`, f_s equals N , so the DFT spacing is exactly 1 Hz. The target PSD is $N_0 = 1e-6$ W/Hz and the positive-frequency bandwidth is 512 Hz. Consequently, the theoretical total power is $5.120000e-04$ W.

The spectral magnitude K is selected through Parseval's relation. After the IFFT, the complex process contains half of its average power in the real part and half in the imaginary part. Multiplication by $\sqrt{2}$ restores the complete target power to each real-valued realization.

3.2 PSD normalization

The script calculates $X = \text{fft}(x1)/N$. The squared magnitude divided by Z gives power in each two-sided DFT bin. Positive-frequency bins are doubled and divided by df to obtain W/Hz. Summing the PSD and multiplying by df gives a spectral estimate of total power that must agree with $\text{mean}(x1.^2)/Z$.

Quantity	Representative result	Interpretation
Theoretical white-noise power	$5.120000e-04$ W	N_0 times 512 Hz
Time-domain estimate	$5.108074e-04$ W	$\text{mean}(x1.^2)/Z$
PSD-integrated estimate	$5.099290e-04$ W	$\text{sum}(\text{PSD}) \cdot df$
Relative time-domain error	-0.23 %	Finite-record fluctuation

3.3 Why the samples look Gaussian

Each time sample is a weighted sum of all spectral phasors. No single phasor dominates because every spectral bin has the same magnitude. The resulting real and imaginary values therefore form nearly Gaussian histograms with zero mean, while the PSD remains approximately uniform across frequency.

4. White-noise results and statistical validation

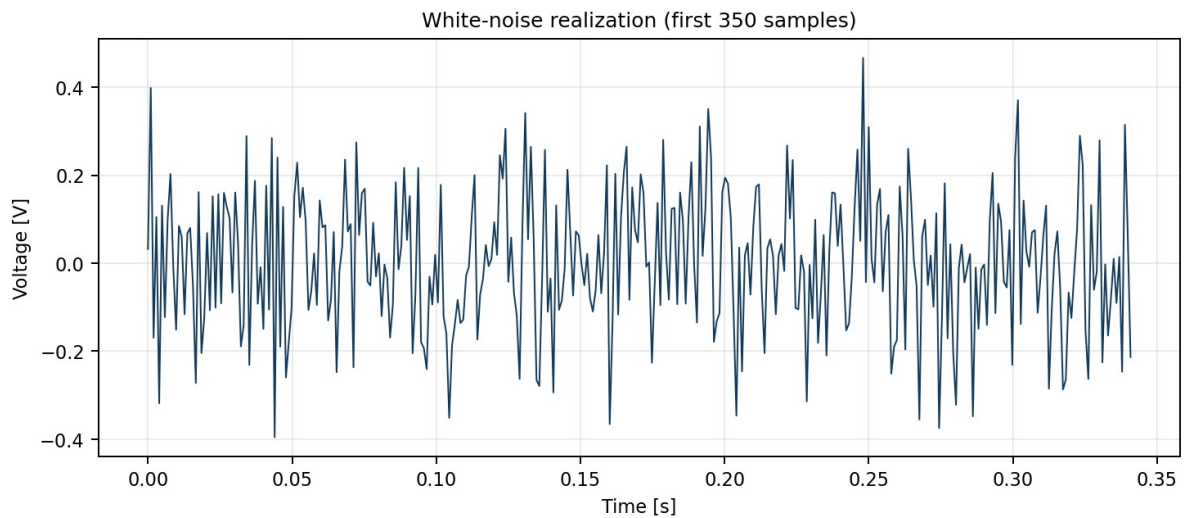


Figure 1. Representative real component of the generated complex noise.

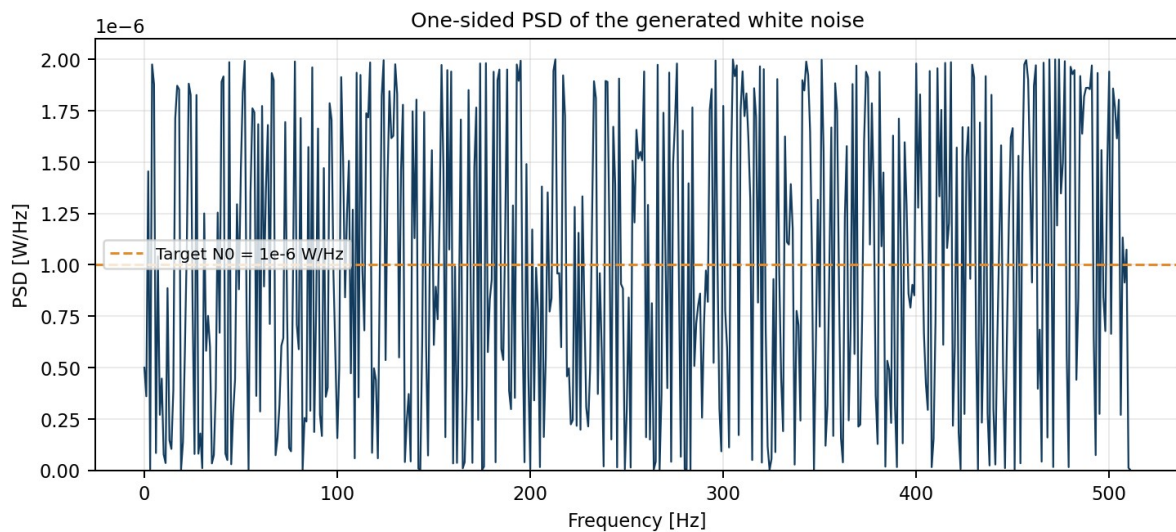


Figure 2. One-sided PSD. Individual bins fluctuate, but the average level follows the target N_0 .

The waveform has no visible periodic structure and rapidly changes sign. In the spectrum, the random bin-to-bin variation is expected for a single periodogram. Averaging several independent records would reduce this variance without changing the mean PSD level.

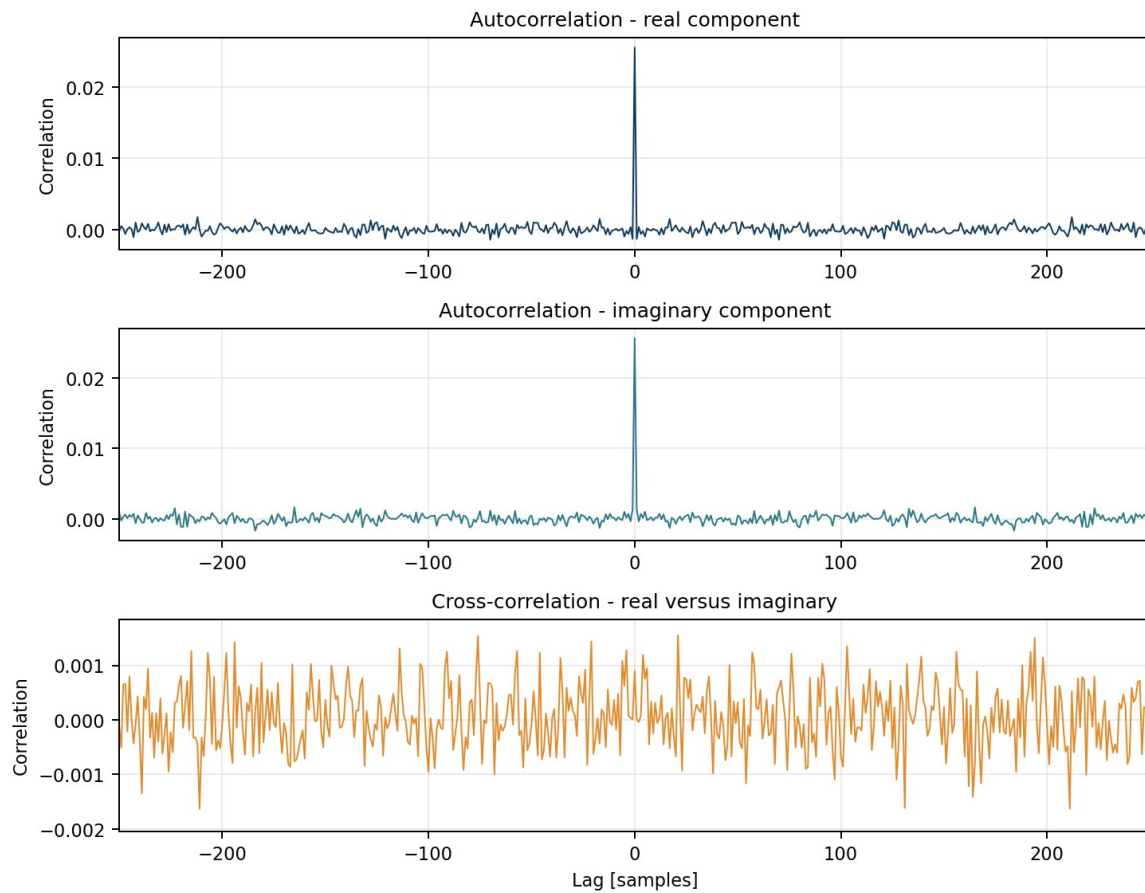


Figure 3. Auto- and cross-correlation estimates for the two real realizations.

The autocorrelation peaks at zero lag and is small elsewhere. The cross-correlation remains centered around zero, supporting the interpretation that the real and imaginary components behave as independent realizations of the same white process.

5. FIR spectral colouring

5.1 Filter design

The first script uses `fir1` to design a 30th-order low-pass FIR filter with a Hamming window. The cutoff frequency is $f_c = N/8 = 128$ Hz, corresponding to a normalized cutoff of 0.25 relative to the Nyquist frequency. The filter has a finite transition band, so its effective noise bandwidth is not exactly equal to 128 Hz.

$$P_{\text{out, expected}} = N_0 * \int_0^{f_s/2} |H(f)|^2 df$$

Metric	Representative result
FIR equivalent noise bandwidth	115.385 Hz
Expected output power	1.153845e-04 W
Estimated output power	1.039191e-04 W
Relative power error	-9.94 %

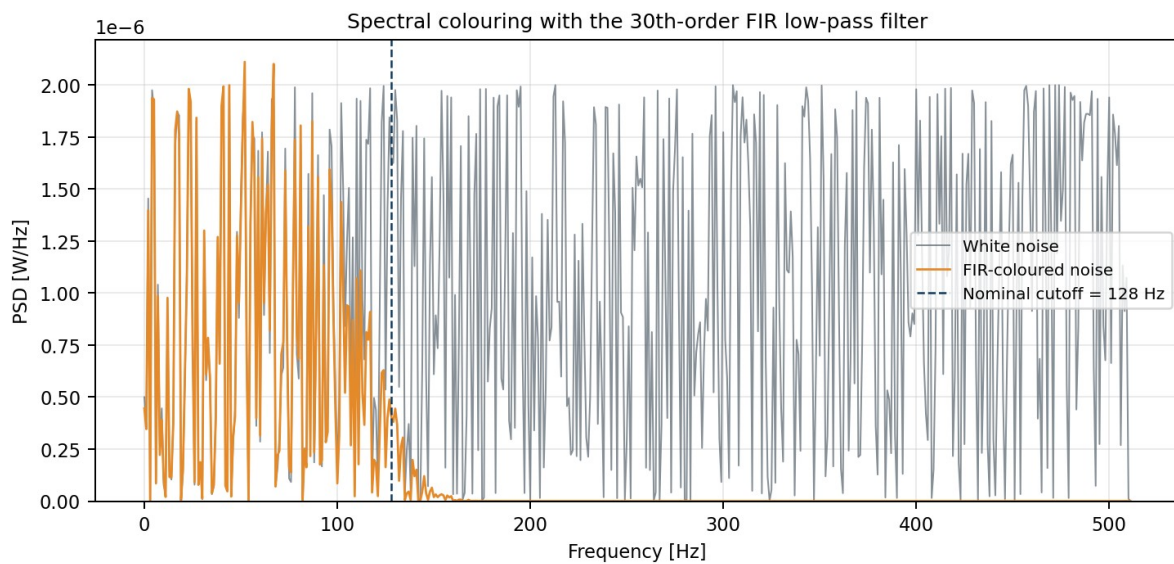


Figure 4. The FIR filter suppresses the high-frequency portion of the white spectrum.

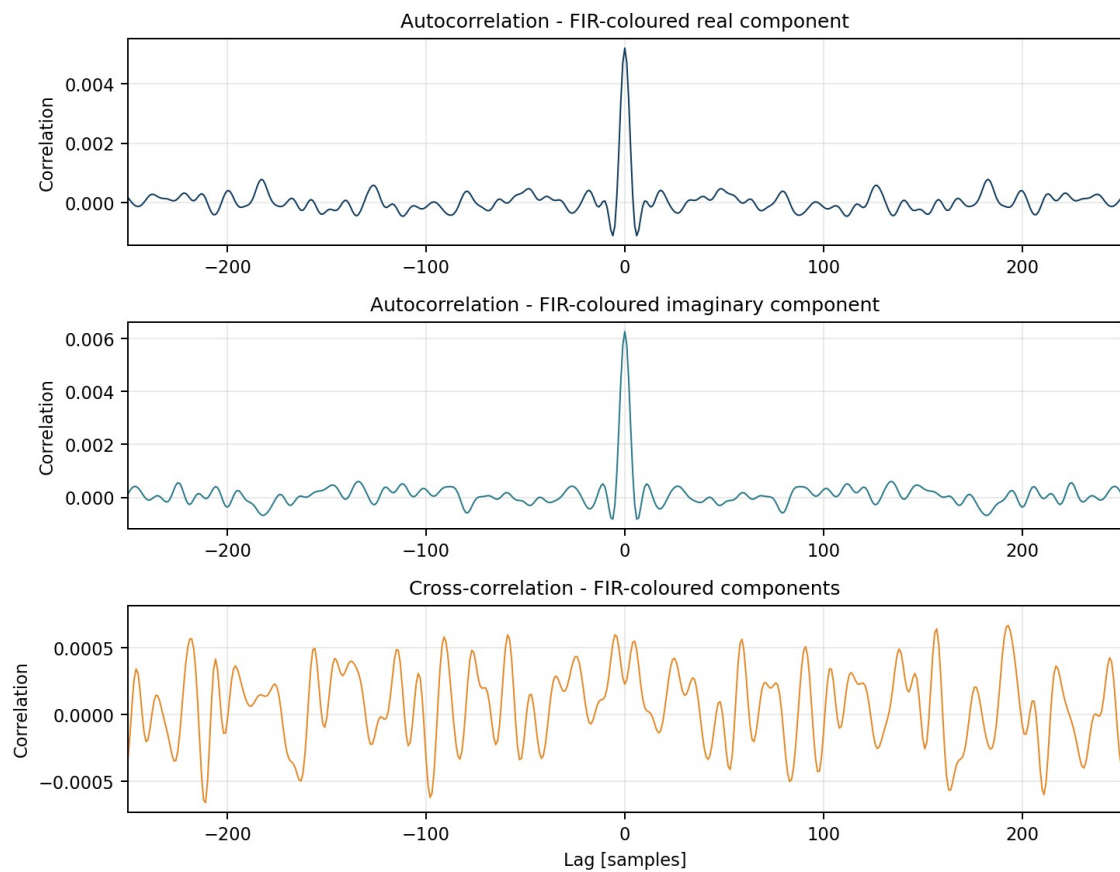


Figure 5. Low-pass colouring broadens the autocorrelation around zero lag.

The output samples are correlated because every sample is a weighted combination of the current and previous input samples. The wider autocorrelation is therefore the time-domain signature of the narrower frequency-domain spectrum.

6. Band-pass coloured noise: Butterworth and FFT mask

6.1 Butterworth IIR filter

RuidoColoreado.m uses a fourth-order Butterworth band-pass design with normalized edges $[1/3, 2/3]$. Since these values are normalized to the Nyquist frequency, the physical passband extends from $f_s/6$ to $f_s/3$, or from 85.333 Hz to 170.667 Hz for $f_s = 512$ Hz.

The recursive Butterworth filter provides a smooth magnitude response and no passband ripple. Its transition bands are gradual, and the impulse response is theoretically infinite. This makes it computationally efficient but not an ideal rectangular selector.

6.2 Ideal FFT mask

The second implementation multiplies the random spectrum by a binary mask. Bins inside the positive-frequency band and the conjugate-position negative-frequency band are retained; all other bins are set to zero. The mask therefore creates abrupt spectral edges and a long, oscillatory time-domain correlation.

```
mask[k] = 1 for fmin <= |f_k| < fmax; otherwise mask[k] = 0
```

6.3 Robust bin selection

The original exercise manually concatenated six vectors of zeros and ones. The delivered script instead constructs the mask from the FFT frequency vector. This removes fragile hard-coded segment lengths and automatically keeps the positive and negative bands symmetric for the selected N and f_s .

The ideal mask is useful as a reference model, but an abrupt rectangular response is noncausal when interpreted as a real-time filter. The Butterworth structure is the practical causal implementation.

7. Band-pass results

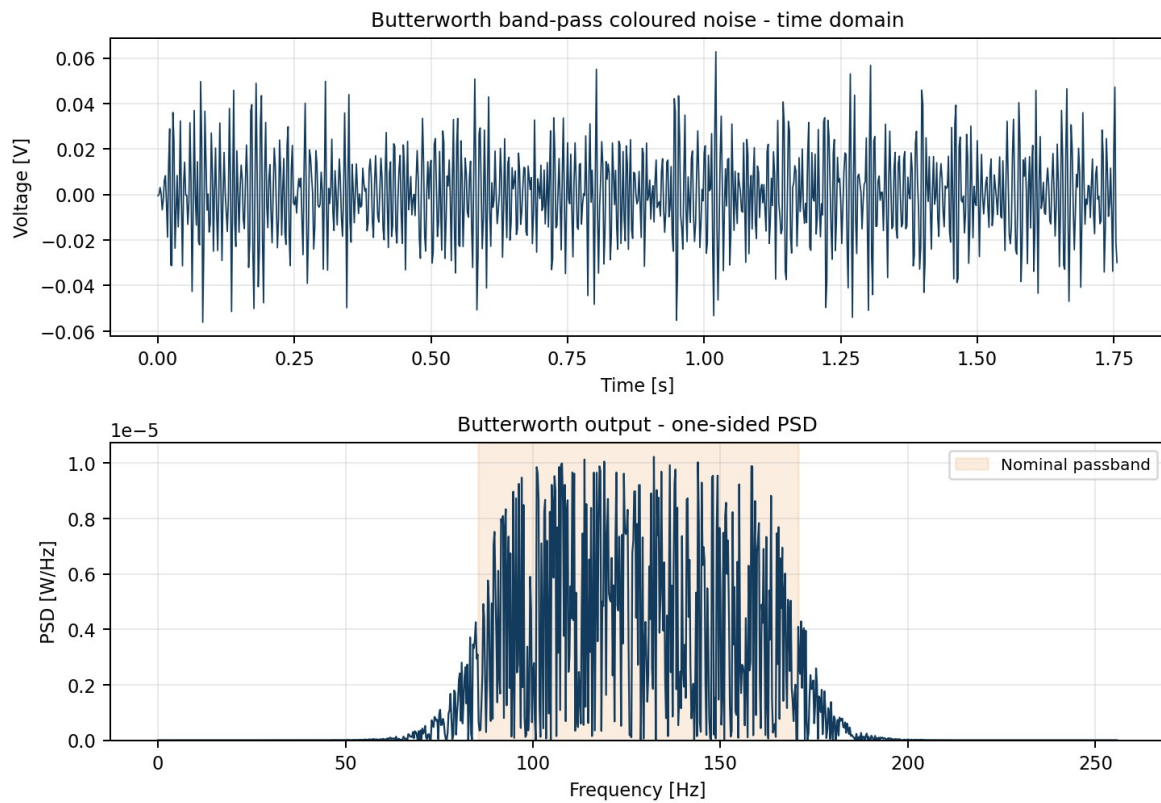


Figure 6. Output of the fourth-order Butterworth band-pass filter.

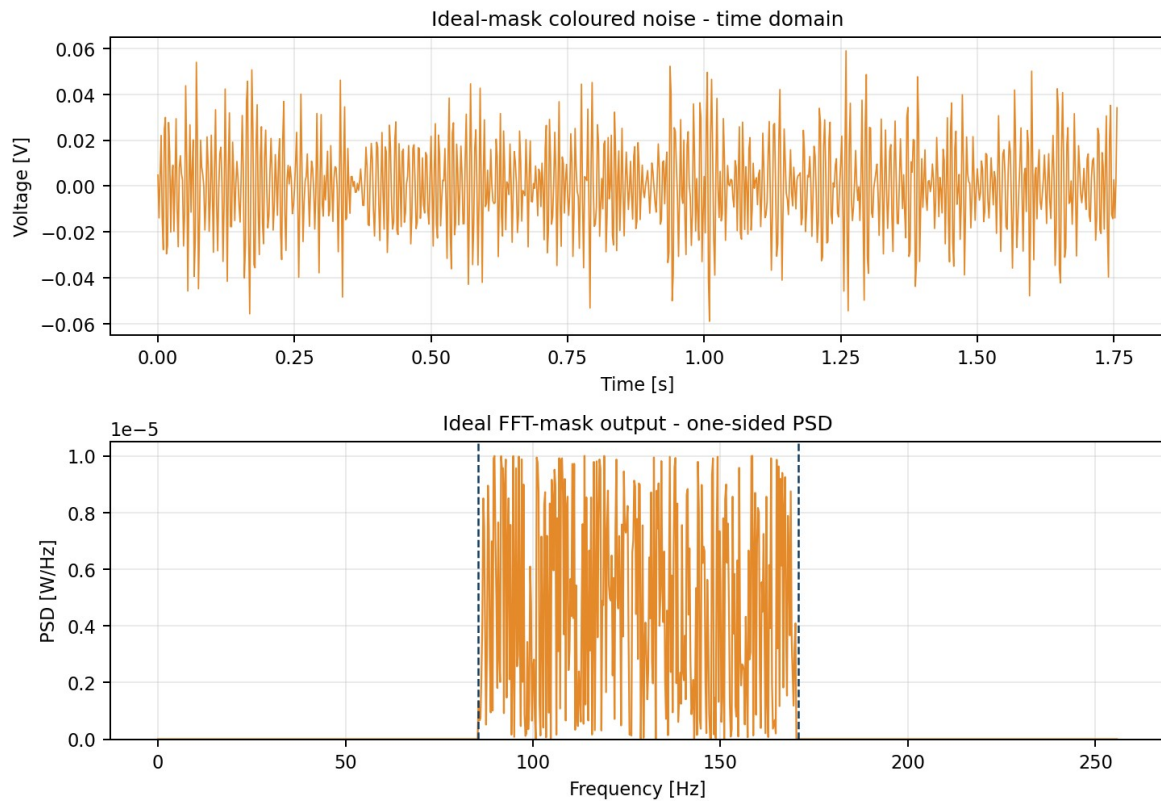


Figure 7. Output obtained by retaining only the ideal passband FFT bins.

Both outputs concentrate their energy inside the same nominal band. The Butterworth response rolls off gradually near the edges, whereas the ideal mask changes abruptly. The time-domain records show an

oscillatory random envelope because a band-pass process can be interpreted as narrowband noise modulating a carrier-like central frequency.

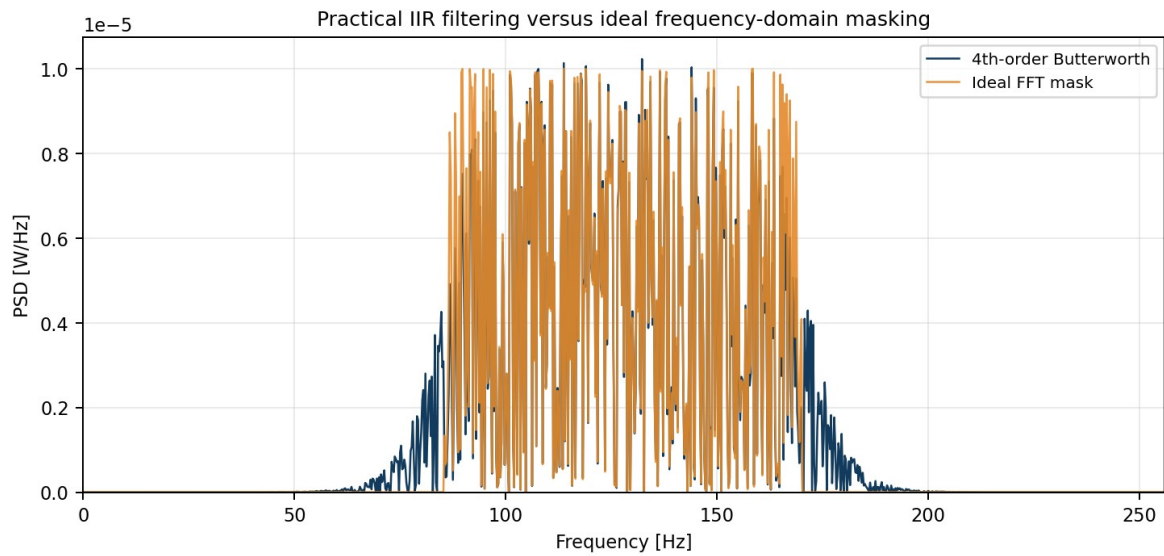


Figure 8. Direct comparison of the practical IIR response and the ideal spectral mask.

8. Power validation and engineering comparison

8.1 Ideal-mask power

The ideal passband width is 85.333 Hz. Multiplication by $N_0 = 5 \times 10^{-6}$ W/Hz gives an expected power of 4.266667×10^{-4} W. The representative time-domain estimate is 3.983012×10^{-4} W, an error of -6.65 %. Because the spectral magnitudes are fixed, the integrated mask power is especially stable.

8.2 Butterworth equivalent noise bandwidth

The Butterworth output power should not be calculated from the nominal edge separation alone because the transition regions contribute partial power. Numerical integration of $|H(f)|^2$ gives an equivalent noise bandwidth of 86.189 Hz. The expected output power is therefore 4.309432×10^{-4} W, compared with an estimated 4.095655×10^{-4} W.

Method	Spectral edge	Causality	Correlation	Power model
White random-phase process	Flat over Nyquist band	Block synthesis	Impulse-like	$N_0 \cdot f_s / 2$
FIR low-pass	Finite transition	Causal	Broad positive central lobe	$N_0 \cdot \text{ENBW}$
Butterworth band-pass	Smooth transition	Causal recursive	Damped oscillation	$N_0 \cdot \text{ENBW}$
Ideal FFT mask	Abrupt rectangular band	Noncausal block reference	Long oscillation	$N_0 \cdot (f_{\text{max}} - f_{\text{min}})$

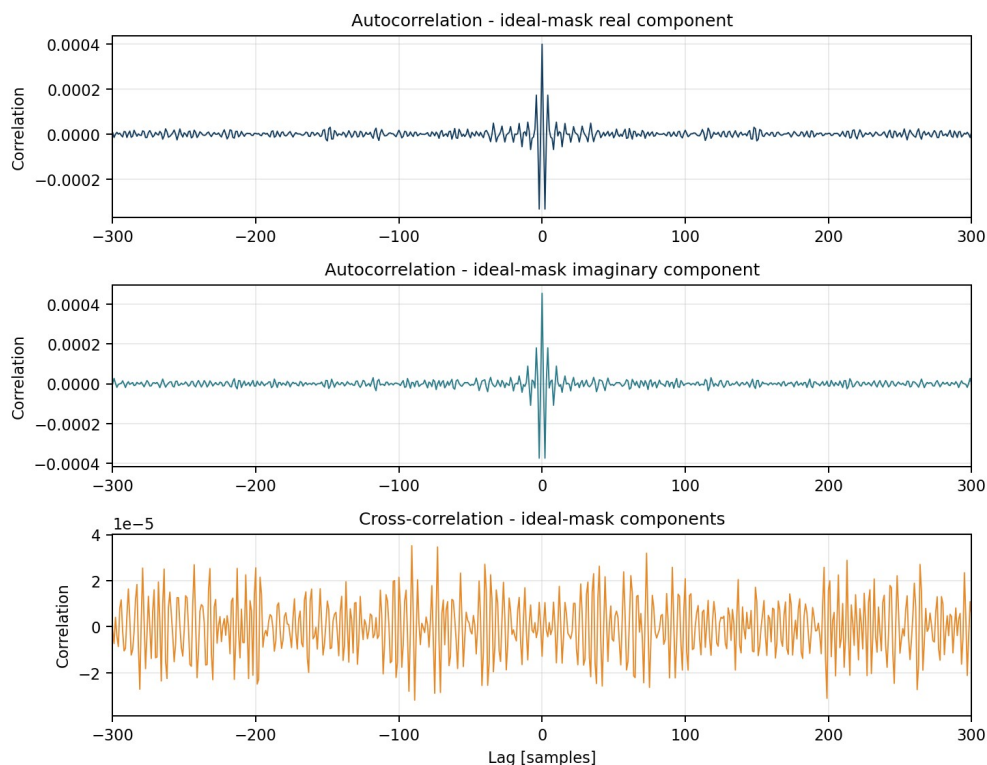


Figure 9. Correlation functions of the ideal-mask band-pass process.

The autocorrelations oscillate because the retained spectrum is centered away from direct current. The slowly decaying envelope is caused by the sharp spectral edges. Cross-correlation between the two extracted real processes remains close to zero.

9. Package organization and execution

9.1 Folder structure

```
Noise_Generation_Practice/  
    README.txt  
    documentation/  
Noise_Generation_MATLAB_Report.pdf  
    codes/  
        RuidoBlancoGaussiano.m  
        RuidoBlancoGaussiano.mat  
        RuidoColoreado.m  
        RuidoColoreado.mat  
    figures/  
figure_01_white_time.png ... figure_09_mask_correlations.png
```

9.2 Difference between .m and .mat

The .m files are executable MATLAB scripts. A .mat file is normally a binary data container rather than source code. Because the requested package mentions code in .mat format, each companion .mat file stores the complete corresponding script in a character variable named code. Load it with `load("RuidoBlancoGaussiano.mat")` and inspect or write the variable as required.

9.3 Recommended execution order

- Set the MATLAB current folder to codes.
- Run `RuidoBlancoGaussiano.m` and inspect Figures 1-6 and the Command Window power values.
- Run `RuidoColoreado.m` and inspect Figures 1-4 and the power comparison.
- Use a different rng seed only when a new independent realization is required.

9.4 Main corrections and improvements

- All comments, titles and variable explanations are written in English.
- The one-sided spectra are divided by df , so the displayed unit is W/Hz rather than power per FFT bin.
- The positive-frequency bins are correctly doubled except at direct current.
- Biased correlation estimates use a consistent normalization for every lag.
- The FFT mask is constructed from frequency conditions rather than hard-coded vector lengths.
- Equivalent noise bandwidth is used for physically correct filtered-noise power prediction.
- Fixed random seeds make the practice reproducible.

10. Conclusions

The practice demonstrates that a white Gaussian-like process can be synthesized efficiently by assigning independent random phases to a constant-magnitude spectrum and applying the inverse FFT. Correct scaling through Parseval's relation and the reference impedance produces the required total electrical power. The real and imaginary components form two nearly independent real-valued noise realizations.

The FFT-based periodogram confirms a flat mean spectral density, while the correlation estimates show the expected impulse-like behavior of white noise. After low-pass FIR filtering, the PSD is confined to low frequencies and the autocorrelation broadens. Frequency-domain concentration and time-domain correlation are therefore two equivalent descriptions of noise colour.

The band-pass experiment compares a causal fourth-order Butterworth implementation with an ideal rectangular FFT mask. Both select the same nominal band, but the IIR transition is smooth and its expected power must be calculated with equivalent noise bandwidth. The ideal mask has exact edges and power NO times the retained bandwidth, but it represents block processing rather than a realizable causal filter.

10.1 Main numerical results

- Script 1 theoretical white-noise power: $5.120000e-04$ W.
- Script 1 representative estimated white-noise power: $5.108074e-04$ W.
- FIR equivalent noise bandwidth: 115.385 Hz; estimated output power: $1.039191e-04$ W.
- Ideal band-pass width: 85.333 Hz; expected power: $4.266667e-04$ W.
- Butterworth equivalent noise bandwidth: 86.189 Hz; representative estimated power: $4.095655e-04$ W.

The exact waveform and individual spectral bins depend on the random seed, but the PSD level, integrated power and correlation structure remain statistically consistent with the theoretical model.

Appendix A. Essential MATLAB fragments

A.1 Random-phase spectrum

```
K = sqrt(Pn*Z*N);
S = K*exp(1i*2*pi*rand(1,N));
x = ifft(S);
x1 = sqrt(2)*real(x);
```

A.2 One-sided PSD in W/Hz

```
X = fft(x1)/N;
Ptwo = abs(X).^2/Z;
Pone = [Ptwo(1), 2*Ptwo(2:N/2)]/(fs/N);
```

A.3 Equivalent noise bandwidth

```
[H,fH] = freqz(b,a,16384,fs);
ENBW = trapz(fH,abs(H).^2);
Pexpected = N0*ENBW;
```

A.4 Symmetric ideal band-pass mask

```
f_fft = (0:N-1)*(fs/N);
mask = ((f_fft>=fmin)&(f_fft<fmax)) | ...
((f_fft>=fs-fmax)&(f_fft<fs-fmin));
y = ifft(S.*mask);
```

A.5 Power checks

```
Ptime = mean(y.^2)/Z;
Pfreq = sum(PSD_one_sided)*(fs/N);
```

The complete, executable versions of these fragments are included in the codes folder. The scripts also generate time-domain plots, PSD plots, correlation diagrams and numerical validation messages.